PROBLEM 19

Heavy Particle on a Paraboloid

Original statement · Problems.pdf, p. 10

Paraboloid Surface

A heavy particle of mass m is moving without friction on the paraboloid which is described as:

$$az = r^2$$

where a is a positive constant and r is the distance to the Oz vertical axis. If $r = a$ initially, and the particle is released with a velocity v_0 tangent to the local parallel, reduce the problem to integrals of the form $\int [...] dr = \int [...] dt$. Determine the maximum and minimum of r along the motion.

Worked Solution (4-Phase Methodology)

PHASE 1

Constraints, Degrees of Freedom & Coordinates

- **System:** a heavy particle of mass m moves without friction on the paraboloid of revolution $az = r^2$, with $a > 0$ and $r = \sqrt{x^2 + y^2}$.
- **Constraint:** $f(x, y, z) = x^2 + y^2 - az = 0$ (holonomic, scleronomic, smooth). The particle is inside the bowl; the reaction must press it against the surface ($\nu > 0$), which is verified in 3.3.
- **Degrees of freedom:** $\text{CDOF} = 3 - 1 = 2$, with the cylindrical coordinates (r, θ) and $z = r^2/a$.
- **Initial state:** $r(0) = a$ (so $z(0) = a$) and velocity v_0 tangent to the local parallel: $\mathbf{v}_0 = v_0\, \mathbf{e}_\theta$, i.e. $\dot{r}(0) = 0, \dot{\theta}(0) = v_0/a, \dot{z}(0) = 0$.
- **Parameters:** m [kg], a [m], g [m/s^2], v_0 [m/s].

PHASE 2

Reference Frames & Vector Bases

- **Inertial frame:** $\mathcal{F}_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ with $\mathbf{k}$ vertical upward.
- **Cylindrical basis:** $\mathcal{B}_1 = \{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}\}$, obtained from $\mathcal{B}_0$ by a rotation $\theta(t)$ about $\mathbf{k}$:

$$[{}_0R_1] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

Multiplying out, RR^T has diagonal entries $\cos^2 + \sin^2 = 1$ of the rotation angle (and 1) and off-diagonal entries $\pm(\cos\sin - \sin\cos) = 0$, so it is the identity.

- **Surface-adapted basis.** With $s = \sqrt{1 + 4r^2/a^2}$, the unit tangent along the meridian is $\mathbf{e}_g = \left(\mathbf{e}_r + \frac{2r}{a}\mathbf{k}\right)/s$ and the unit normal pointing into the bowl is $\mathbf{n}_{in} = \left(\mathbf{k} - \frac{2r}{a}\mathbf{e}_r\right)/s$, proportional to $-\nabla f = -2r\,\mathbf{e}_r + a\,\mathbf{k}$. Since $\mathbf{e}_g \times \mathbf{e}_\theta = \mathbf{n}_{in}$ the triple is right-handed, and

$$[{}_1R_2] = \begin{pmatrix} 1/s & 0 & -2r/(as) \\ 0 & 1 & 0 \\ 2r/(as) & 0 & 1/s \end{pmatrix}, \qquad \det[{}_1R_2] = \frac{1}{s^2} + \frac{4r^2}{a^2 s^2} = 1, \qquad [{}_1R_2][{}_1R_2]^T = I$$

PHASE 3

Derivation

3.1 Kinematics

Pedagogical Justification: the constraint is used to write $z = r^2/a$ and its derivatives with the chain rule, so that only (r, θ) remain.

$$z = \frac{r^2}{a}, \qquad \dot{z} = \frac{2r\dot{r}}{a}, \qquad \ddot{z} = \frac{2}{a}\left(\dot{r}^2 + r\ddot{r}\right)$$

$$\mathbf{v} = \dot r\,\mathbf{e}_r + r\dot\theta\,\mathbf{e}_\theta + \frac{2r\dot r}{a}\,\mathbf{k}, \qquad v^2 = \dot r^2\left(1+\frac{4r^2}{a^2}\right) + r^2\dot\theta^2$$

$$\mathbf{a} = \left(\ddot r - r\dot\theta^2\right)\mathbf{e}_r + \left(r\ddot\theta + 2\dot r\dot\theta\right)\mathbf{e}_\theta + \ddot z\,\mathbf{k}$$

3.2 Forces

- **Applied force:** the weight $-mg\,\mathbf{k}$, with $V = mgz = mgr^2/a$.
- **Constraint force:** the reaction of the smooth surface along $\mathbf{n}_{in}$ (no $\mathbf{e}_\theta$ component, because $\nabla f = 2r\,\mathbf{e}_r - a\,\mathbf{k}$ has none): $\mathbf{N} = \nu\,\mathbf{n}_{in} = \dfrac{\nu}{s}\left(\mathbf{k} - \dfrac{2r}{a}\mathbf{e}_r\right)$. **Inertia forces:** none.

3.3 Equations of Motion

> **Pedagogical Justification:** Newton's second law in S_0 gives three scalar equations for the unknowns r, θ and ν:

$$\mathbf{e}_r:\ m\left(\ddot r - r\dot\theta^2\right) = -\frac{2r}{as}\,\nu, \qquad \mathbf{e}_\theta:\ m\left(r\ddot\theta + 2\dot r\dot\theta\right) = 0, \qquad \mathbf{k}:\ m\ddot z = \frac{\nu}{s} - mg$$

- **Decoupling.** Eliminating ν/s between the $\mathbf{e}_r$ and $\mathbf{k}$ equations gives one second-order equation for r (the reaction is computed afterwards from the $\mathbf{k}$ equation, see below).
- **Reaction.** From the $\mathbf{k}$ equation $\nu/s = m(\ddot z + g)$. The $\mathbf{e}_r$ equation gives $\ddot r - r\dot\theta^2 = -\frac{2r}{a}(\ddot z + g)$; with $\ddot z = \frac{2}{a}(\dot r^2 + r\ddot r)$ this is $r\ddot r\,s^2 = r^2\dot\theta^2 - \frac{4r^2\dot r^2}{a^2} - \frac{2gr^2}{a}$, hence $\dot r^2 + r\ddot r = \left(\dot r^2 + r^2\dot\theta^2 - \frac{2gr^2}{a}\right)/s^2$ and, using $gs^2 - 4gr^2/a^2 = g$,

$$\ddot z + g = \frac{g + \frac{2}{a}\left(\dot r^2 + r^2\dot\theta^2\right)}{s^2} \;\Longrightarrow\; \nu = \frac{m}{s}\left[g + \frac{2}{a}\left(\dot r^2 + r^2\dot\theta^2\right)\right] > 0$$

so the surface always pushes the particle and the motion is admissible for any v_0.

- **First integral from $\mathbf{e}_\theta$.** As in Problem 18, $\frac{1}{r}\frac{d}{dt}\left(r^2\dot\theta\right) = 0$, so

$$r^2\dot\theta = r(0)^2\dot\theta(0) = a^2\cdot\frac{v_0}{a} = av_0 \implies \dot\theta = \frac{av_0}{r^2}$$

- **Energy.** The reaction does no work ($\mathbf{N}\cdot\mathbf{v}=0$) and gravity is conservative, so $E = \frac{1}{2}mv^2 + mg\dfrac{r^2}{a}$ is constant. At $t=0$: $E = \frac{1}{2}mv_0^2 + mga$.

3.4 Integration and Answers

Reduction to quadratures

> **Pedagogical Justification:** substitute $\dot\theta = av_0/r^2$ in the energy and multiply by $2/m$:

$$\dot r^2\left(1+\frac{4r^2}{a^2}\right) + \frac{a^2v_0^2}{r^2} + \frac{2gr^2}{a} = v_0^2 + 2ga \quad\Longleftrightarrow\quad \dot r^2\left(1+\frac{4r^2}{a^2}\right) + r^2\left(\dot\theta^2 + \frac{2g}{a}\right) = v_0^2 + 2ga$$

- Isolating $\dot r$, taking the square root and separating variables (Barrow's rule from $r(0)=a$, $t=0$):

$$\dot r^2 = \frac{v_0^2 + 2ga - a^2v_0^2/r^2 - 2gr^2/a}{1+4r^2/a^2} \implies \int_a^r \sqrt{\frac{1+4r'^2/a^2}{v_0^2+2ga-a^2v_0^2/r'^2-2gr'^2/a}}\,dr' = \pm\int_0^t dt' = \pm t$$

with $\theta(t) = \displaystyle\int_0^t \frac{av_0}{r(t')^2}\,dt'$.

Maximum and minimum of r

> **Pedagogical Justification:** the turning points satisfy $\dot r = 0$. Multiplying the numerator by $-ar^2$ and writing $u = r^2$:

$$2gu^2 - a\left(v_0^2 + 2ga\right)u + a^3v_0^2 = 0$$

- Since the particle starts at $r=a$ with $\dot r = 0$, $u = a^2$ is a root: $2ga^4 - a^3v_0^2 - 2ga^4 + a^3v_0^2 = 0$. Factorising:

$$2gu^2 - a\left(v_0^2 + 2ga\right)u + a^3 v_0^2 = \left(u - a^2\right)\left(2gu - av_0^2\right) = 0 \implies r_1 = a, \qquad r_2 = v_0\sqrt{\frac{a}{2g}}$$

- **Case $v_0 < \sqrt{2ga}$:** $r_2 < a$, so $r_{\max} = a$ and $r_{\min} = v_0\sqrt{a/(2g)}$. The particle falls toward the vertex, reaches $r_{\min}$ and climbs back to a.
- **Case $v_0 = \sqrt{2ga}$:** $r_2 = a$, a double root: $r_{\min} = r_{\max} = a$, a horizontal circular orbit.
- **Case $v_0 > \sqrt{2ga}$:** $r_2 > a$, so $r_{\min} = a$ and $r_{\max} = v_0\sqrt{a/(2g)}$. The particle climbs to $r_{\max}$ and falls back to a.

PHASE 4

Interpretation & Checks

- **Circular orbit (case $v_0^2 = 2ga$):** the quadratic becomes $2gu^2 - 4ga^2u + 2ga^4 = 2g\left(u - a^2\right)^2$, a double root, so $\dot{r} \equiv 0$. This agrees with the circular-orbit condition $v^2 = gr\tan\gamma$ with $\tan\gamma = dz/dr = 2r/a = 2$ at $r = a$.
- **Limit $v_0 \to 0$:** $r_{\min} \to 0$, the particle drops to the vertex, as expected for a body released at rest.
- **Dimensional check:** $[v_0^2 + 2ga] = [a^2 v_0^2/r^2] = [2gr^2/a] = \mathrm{m}^2/\mathrm{s}^2$, and $\left[v_0\sqrt{a/(2g)}\right] = \mathrm{m}$.
- **Check against the official key:** the key of `Problems.pdf` gives the quadrature $\dot{r}^2\left(1 + 4r^2/a^2\right) + r^2\left(\dot{\theta}^2 + 2g/a\right) = v_0^2 + 2ga$ with $r^2\dot{\theta} = av_0$, and the extremes: if $v_0 < \sqrt{2ga}$, $r_{\max} = a$ and $r_{\min} = v_0\sqrt{a/(2g)}$; if $v_0 = \sqrt{2ga}$, $r = a$ is constant; if $v_0 > \sqrt{2ga}$, $r_{\min} = a$ and $r_{\max} = v_0\sqrt{a/(2g)}$. All coincide with the results above. Numerical integration of the radial equation ($g = a = 1$) gives the turning points 0.707107 for $v_0 = 1$, 1.000000 for $v_0 = \sqrt{2}$ and 1.414214 for $v_0 = 2$, equal to $v_0\sqrt{a/(2g)}$.

Source: Problems.pdf